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Pedestrian Automatic Emergency Braking (P-AEB)

Alan Moore, P.E.
A.B.Moore Forensic Engineering, Inc.
Orlando, FL

EDR SUMMIT
Research • Collection • Analysis

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Alan Moore, P.E.

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- Mr. Moore is a mechanical engineer and principal of A.B.Moore Forensic Engineering. He specializes in vehicle accident reconstruction, vehicle design analysis, and mechanical engineering consulting.

- Mr. Moore holds a Bachelor of Science in Mechanical Engineering from Michigan State University and a Masters degree in Business Administration from the University of Florida. He is a licensed Professional Engineer, a Board Certified Forensic Engineer, and an ACTAR-certified accident reconstructionist. His past experience includes two decades of accident reconstruction and automotive engineering. He previously worked at Ford Motor Company as a design engineer on the Ford Expedition and Excursion. Mr. Moore also serves as a high-performance driving coach for aspiring race car drivers through the Porsche Club of America.



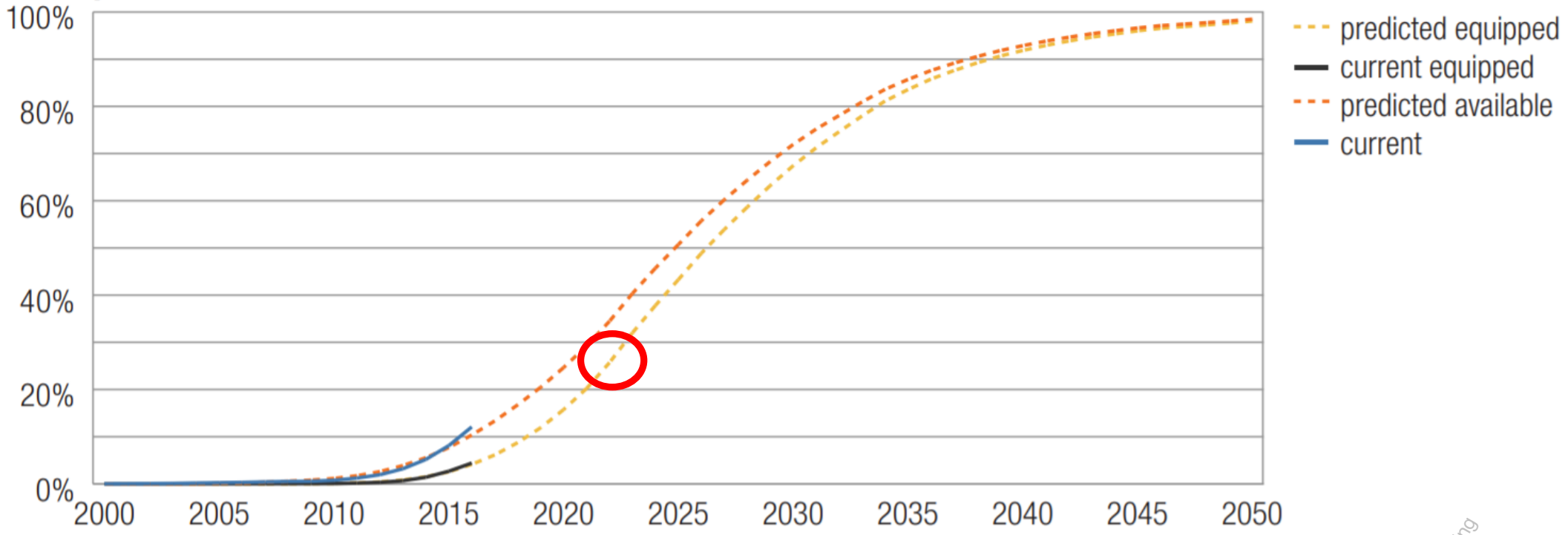
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The Future of ADAS and Accident Reconstruction

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Figure 6: Predicted percentage of registered vehicles with front crash prevention

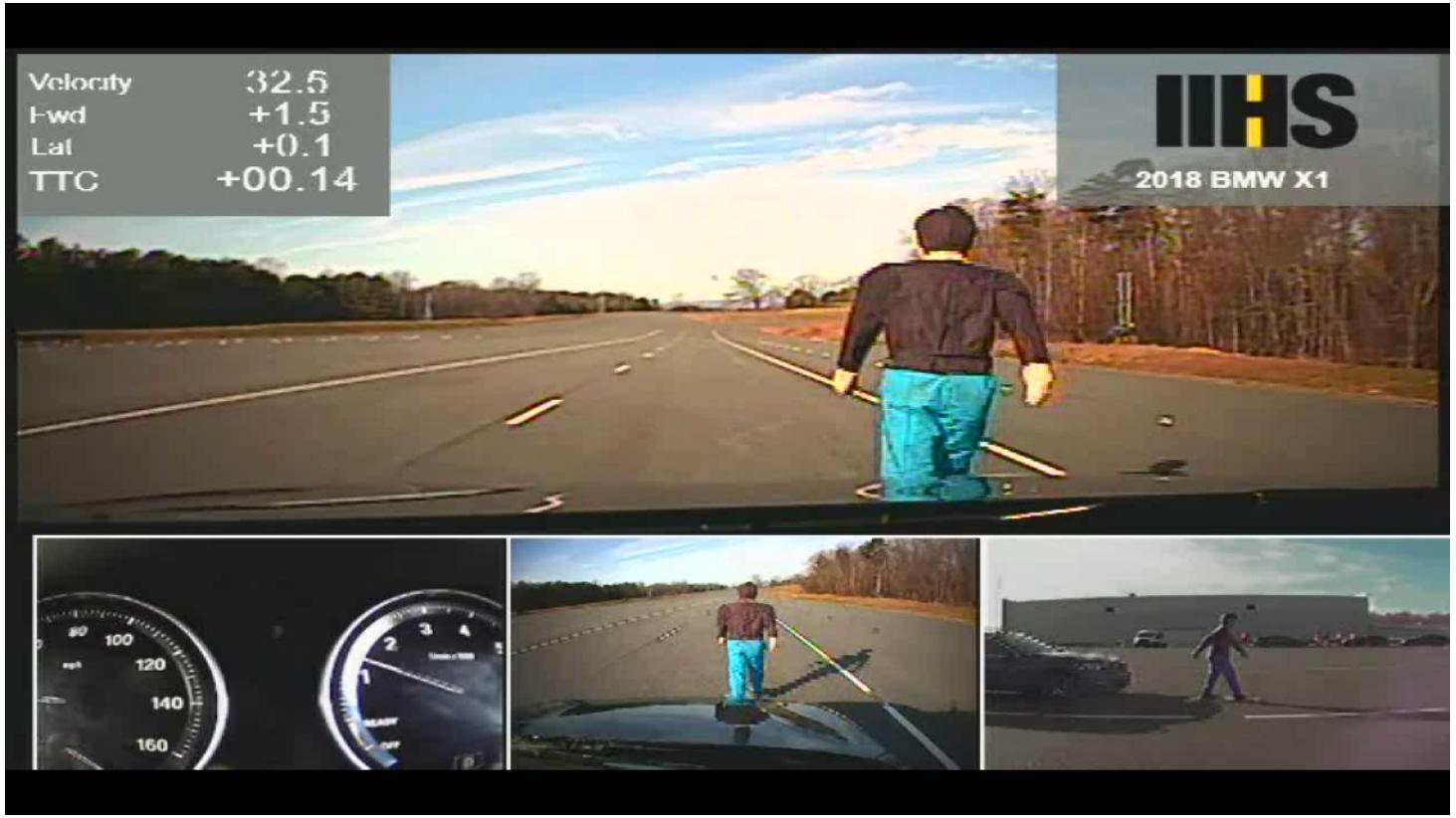


Predicted availability and fitment of safety features on registered vehicles, IIHS/HLDI, 2018

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IIHS Blooper Reel

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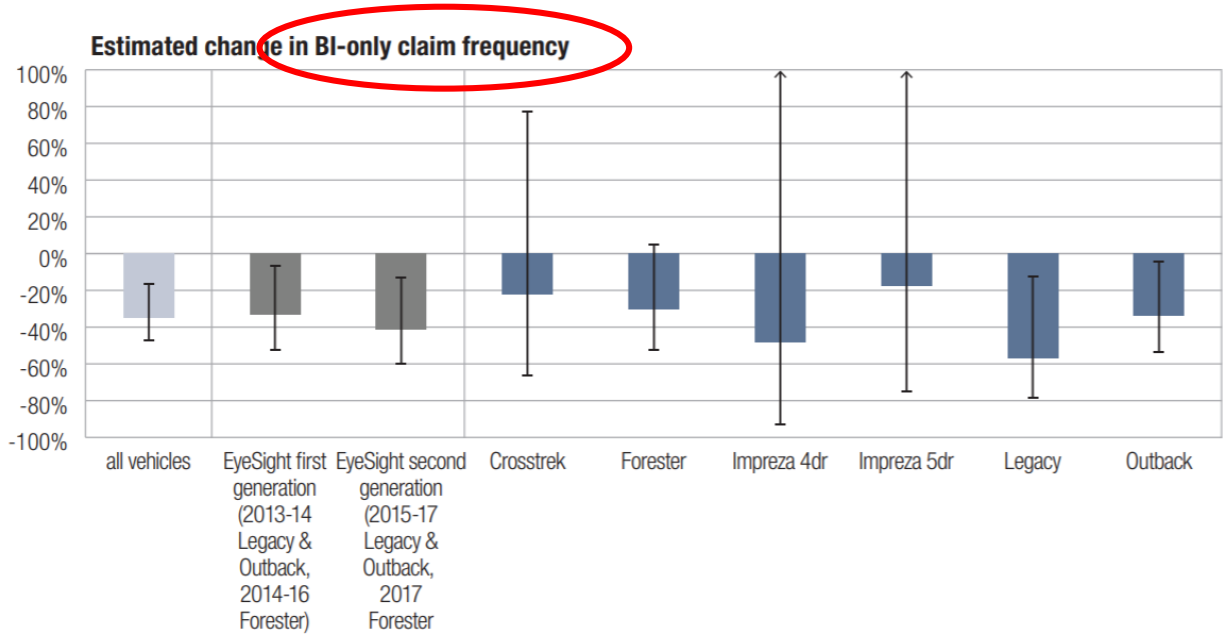
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Effectiveness of Subaru Pedestrian Detection

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Subaru vehicles equipped with EyeSight reduced pedestrian-related claim frequency by 35 percent compared to Subaru vehicles without EyeSight



Effect of Subaru EyeSight on pedestrian-related bodily injury liability claim frequencies, Bulletin Vol. 34, No. 39 : December 2017
https://www.iihs.org/media/e8f617a5-b8f8-42e6-8b0f-9f56f9cb4208/W63Mlg/HLDI%20Research/Bulletins/hldi_bulletin_34.39.pdf

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What is P-AEB?

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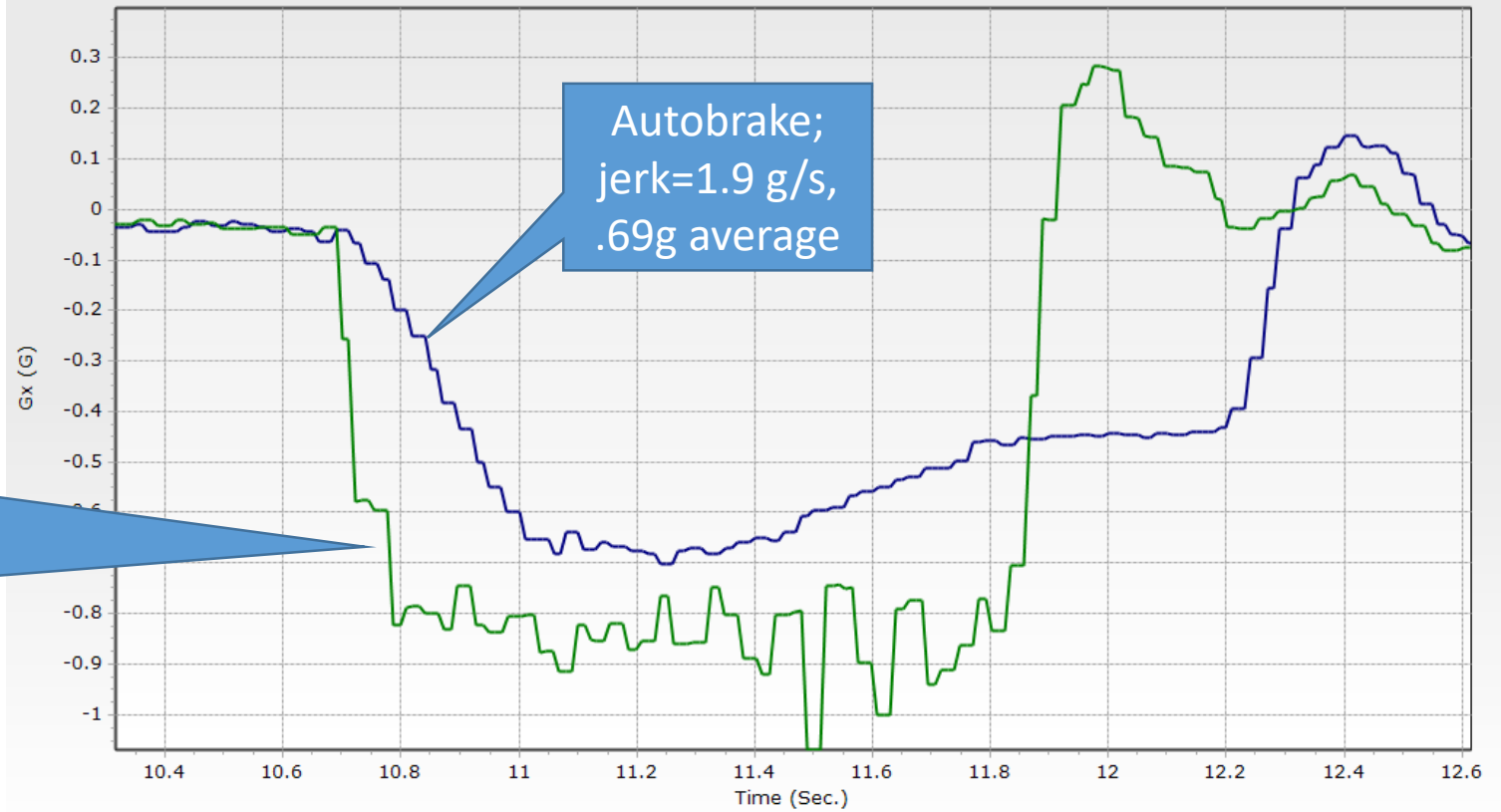
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- Forward Collision Warning
- Brake Assist
- Automatic Emergency Braking
- Balancing risk of false positives and false negatives

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2018 Toyota Camry – Brake Assist v Autobrake

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18mph. 0.9s from warning to autobrake

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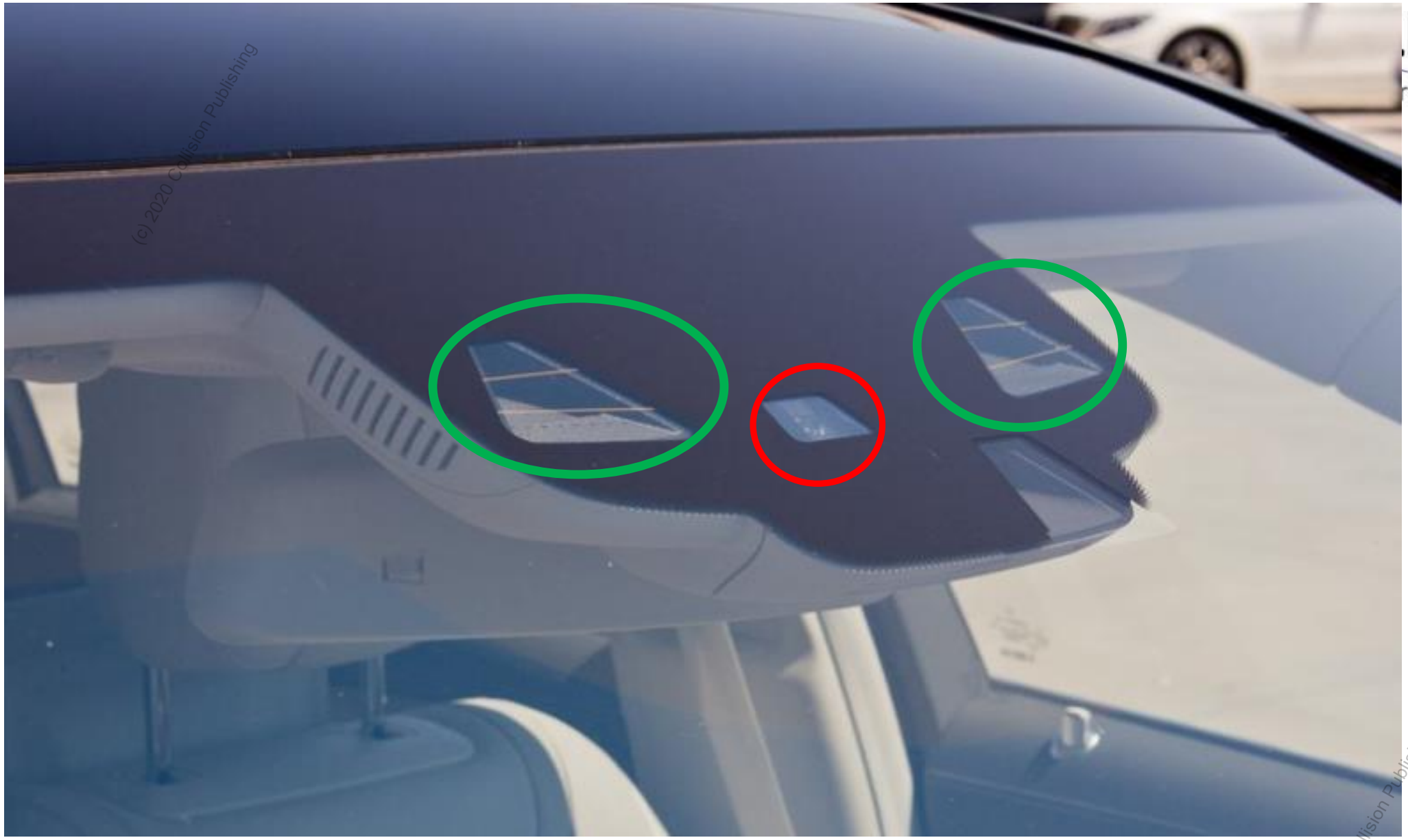
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Pedestrian Automatic Emergency Braking (P-AEB) – How it works

- Sensors
- Image Classification and Computing
- Braking Hardware

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Image Classification

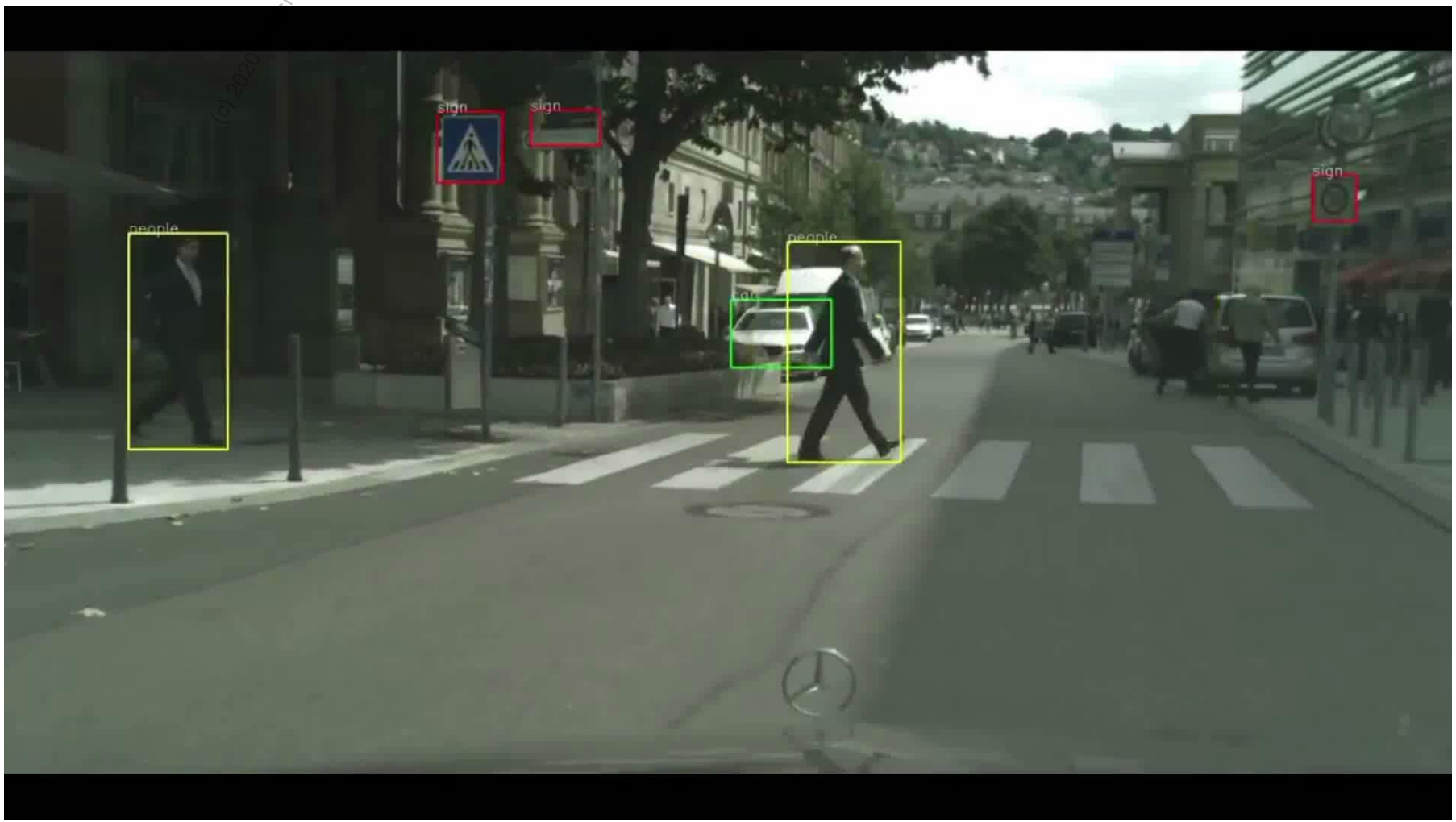
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CES 2016 Nvidia DriveNet demo video



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Tesla Autopilot Visualization

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2019 Ford Edge Pedestrian Detection – Camera Only

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In-Path Stationary Object (IPSO)

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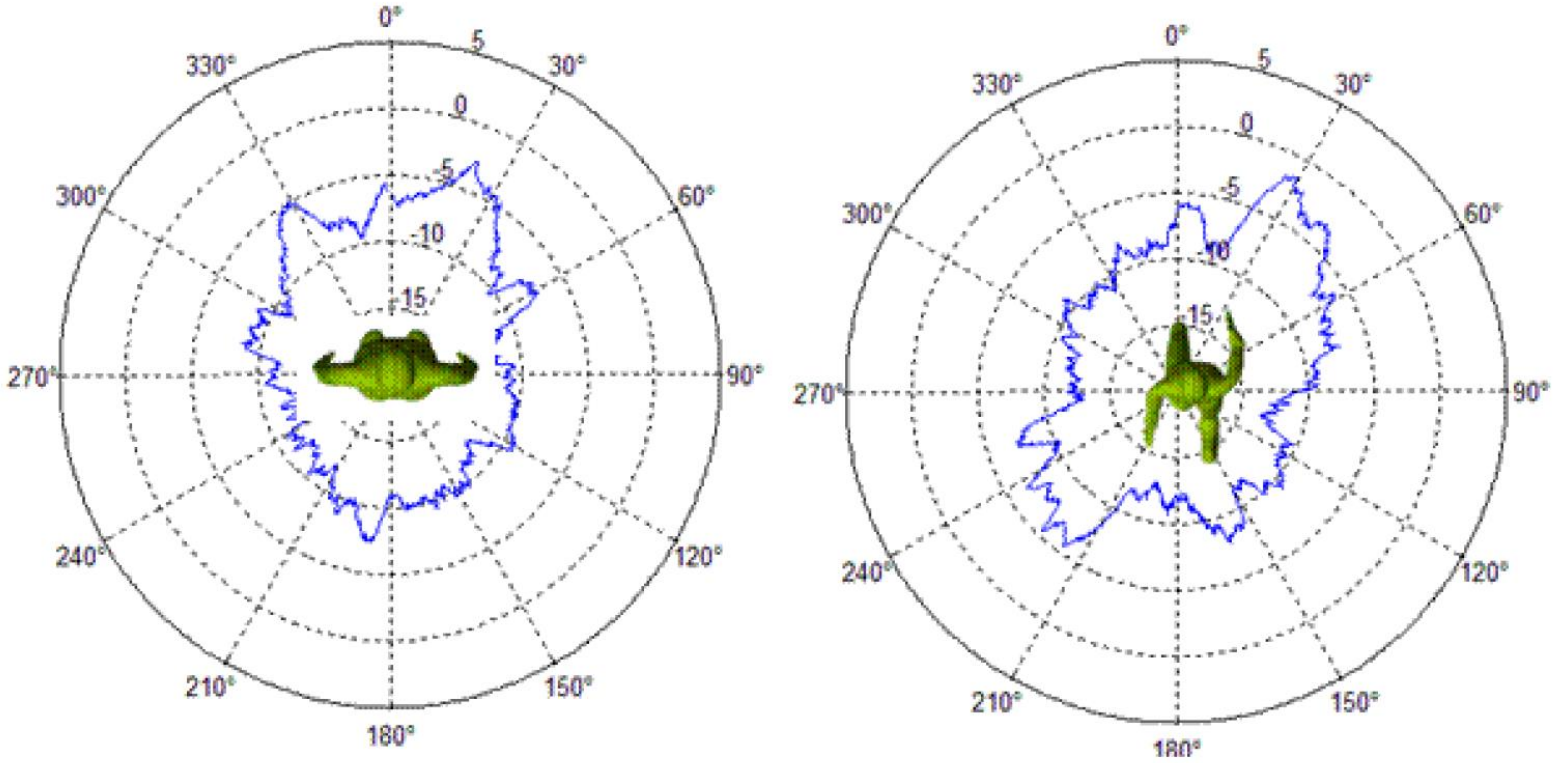
An object is classified as an IPSO if it is associated with a stationary radar target—specific distance measurement from the radar system—and the camera system does not detect a recognized object at that distance, but does detect lack of *free space* or clear path in travel direction of the Tesla. For example, a highway overpass would likely not be classified as IPSO, even though a radar system may detect the overpass as a stationary object, but camera system *would* detect available free space in the vehicle's path underneath the overpass.

NTSB – Human Performance Factors Group Chairman's Supplemental Report: Driver Assistance Systems Williston, FL HWY16FH018

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Radar Pedestrian Detection

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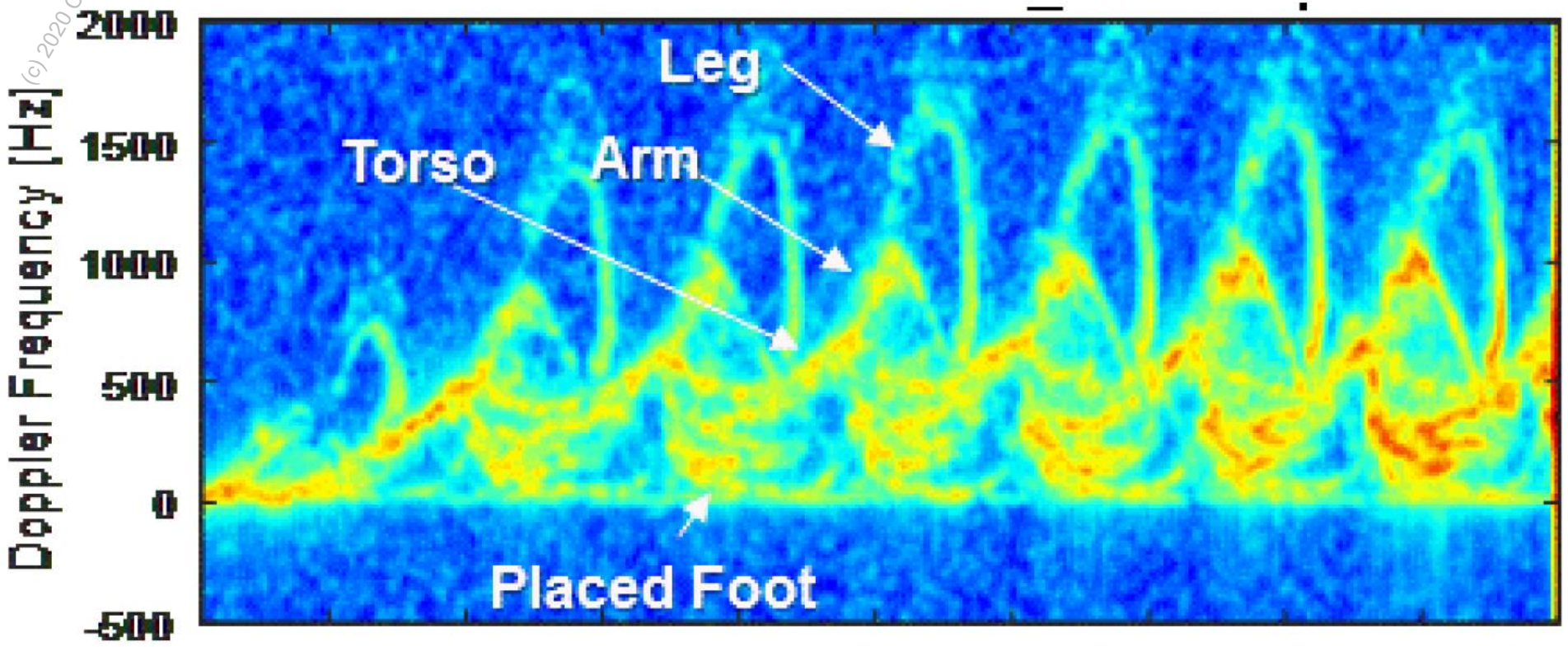


SAE J3116, Surface Vehicle Recommend Practice, June 2017

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Radar Pedestrian Detection

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Micro-Doppler effect of a ped walking toward a 77GHz radar.

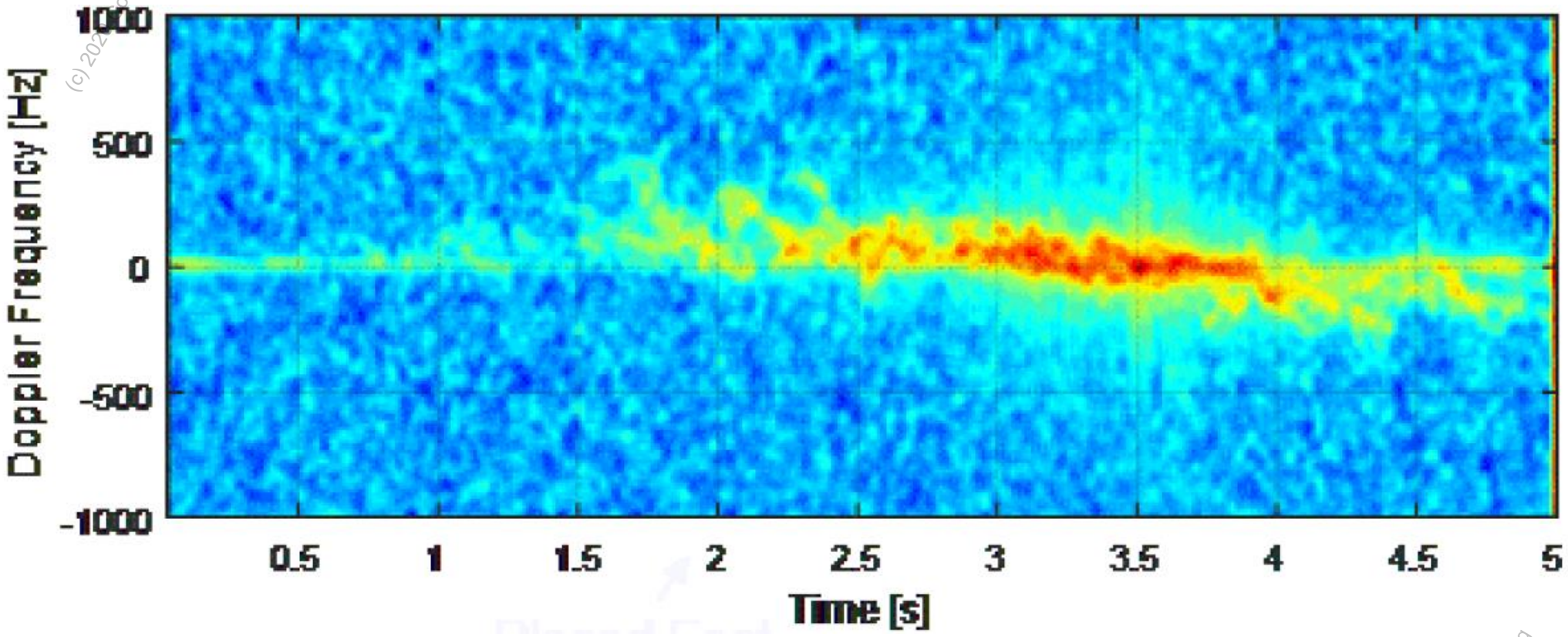
SAE J3116, Surface Vehicle Recommend Practice, June 2017

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Radar Pedestrian Detection

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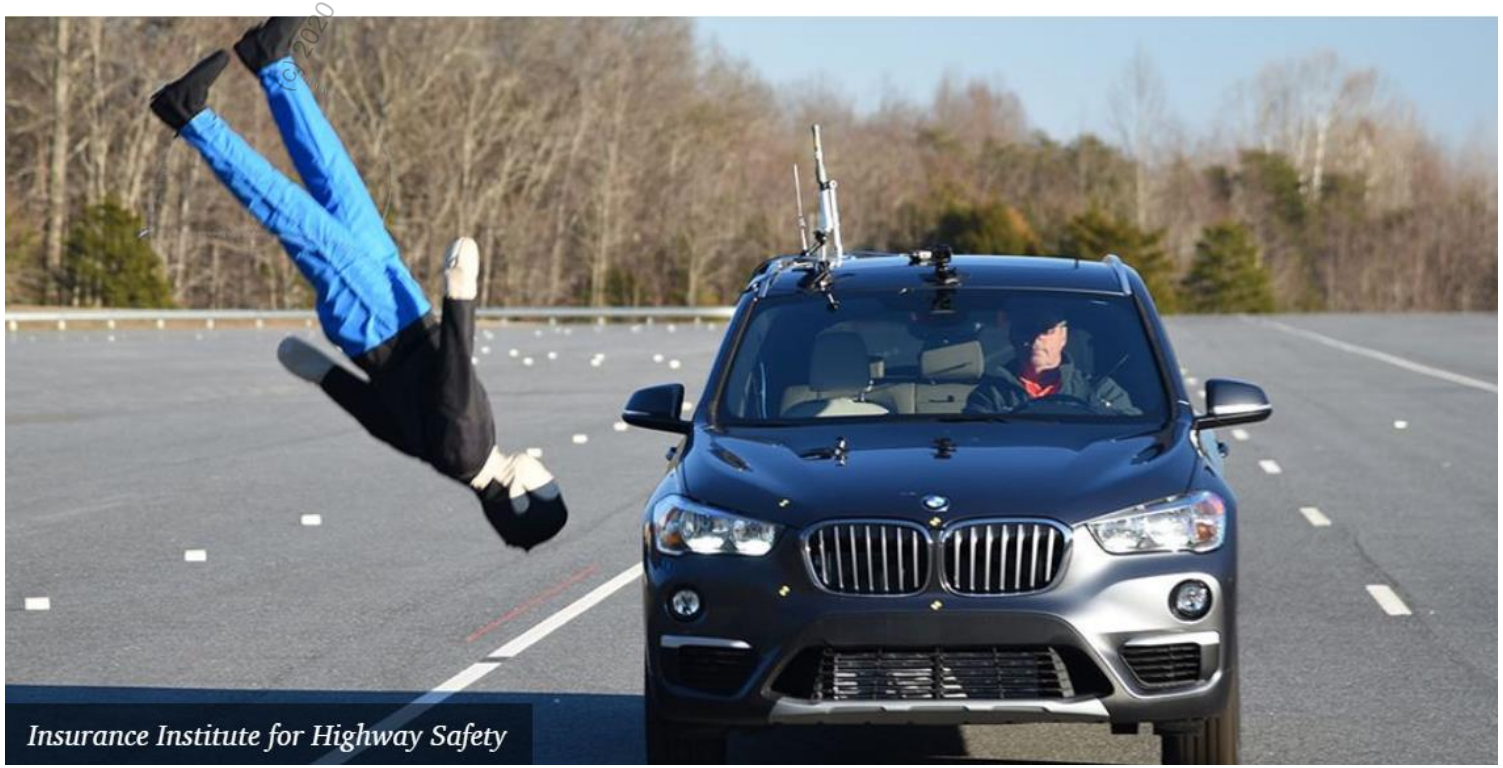
Micro-Doppler effect of a ped walking across a 77GHz radar.

SAE J3116, Surface Vehicle Recommend Practice, June 2017

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Why is Pedestrian Detection Challenging?

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- Short TTC
- Multiple trajectories – perpendicular, parallel, both moving, not in lane
- Sensing limitations
- Image classification
- Balancing risk
- Edge cases

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Why is Pedestrian Detection Challenging?

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Edge cases exist, where the target's speed, gait, radar reflection, and image classification vary.

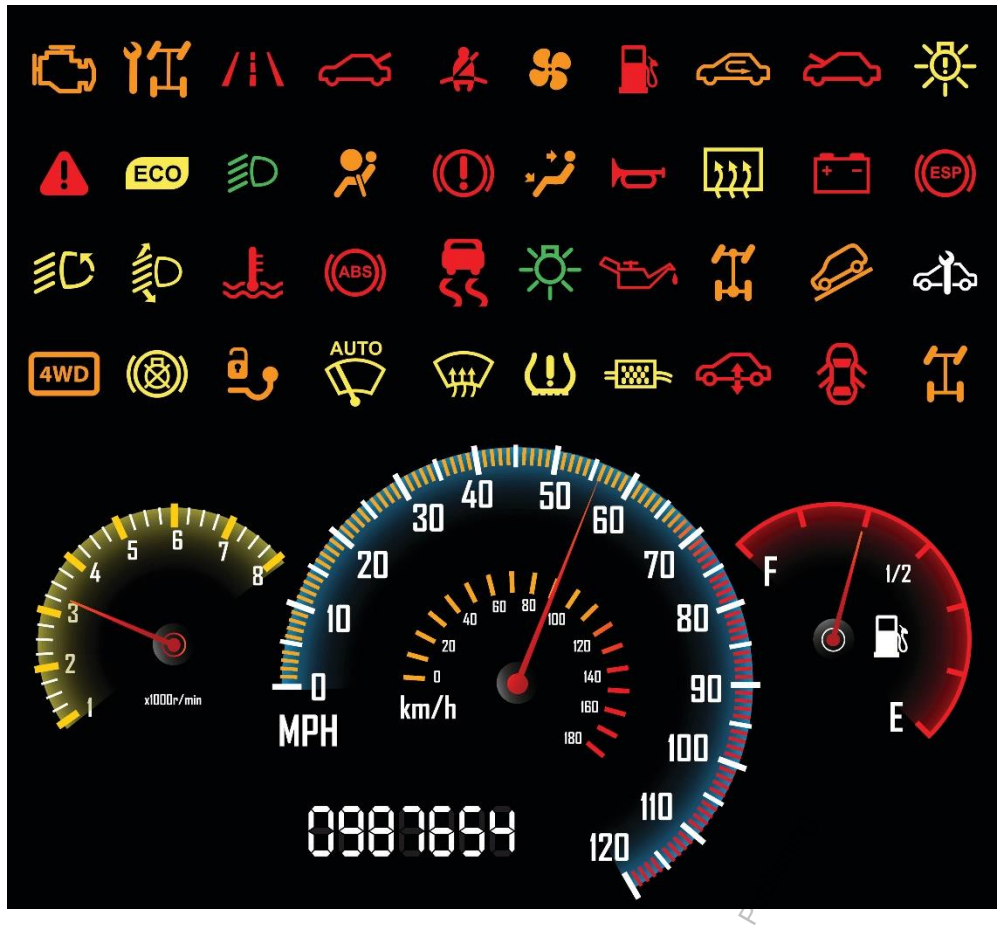
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Driver braking in response to audible/visual alert

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0.8s from known sound to known response (foot on the brake pedal).

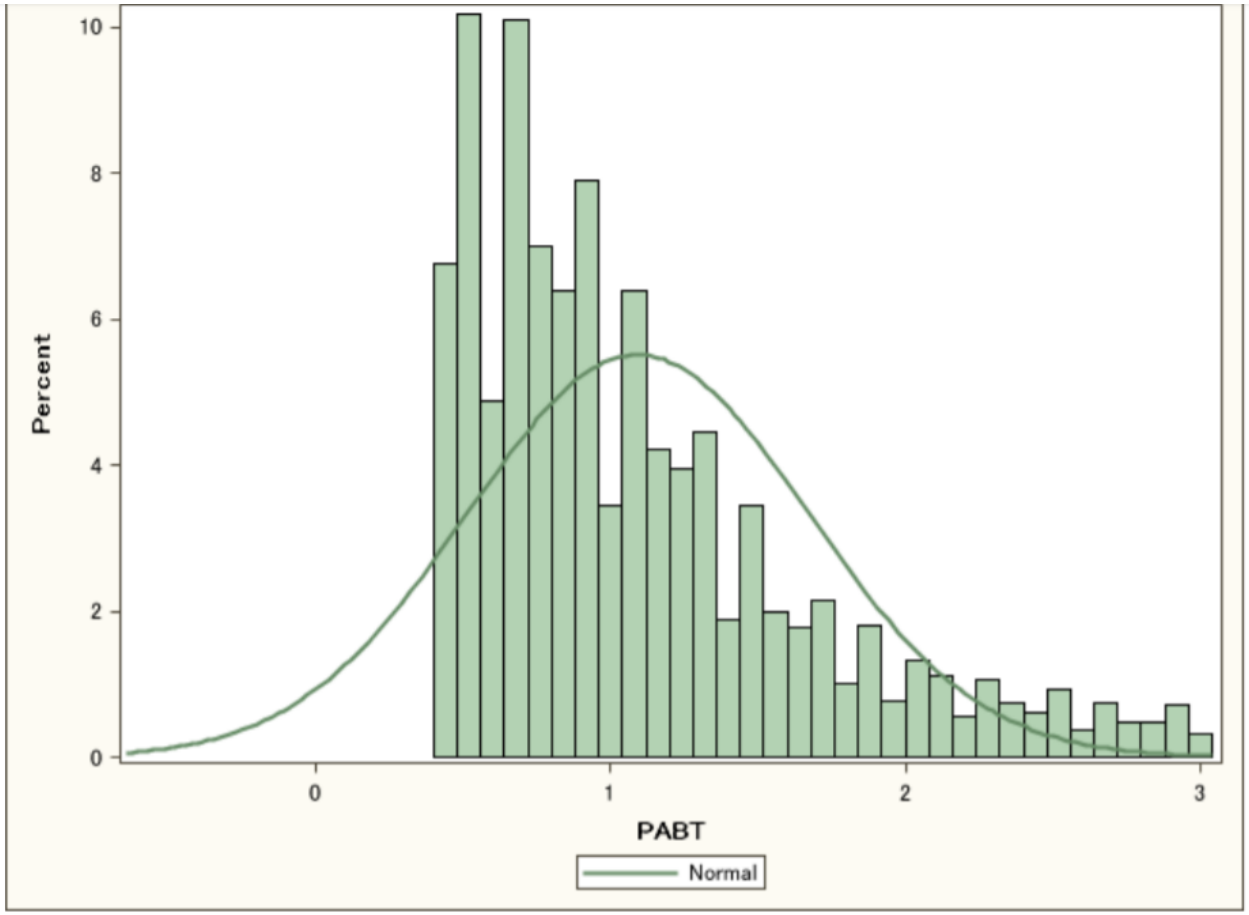


IDRR Software and Related Research, Jeff Muttart and Swaroop Dinakar, 2019

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FCW – Driver Response Time

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Large-Scale Field Test of Forward Collision Alert And Lane Departure Warning Systems, DOT HS 812 247, 2016
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812247b-lanedeparturewarningsystems.pdf

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Driver Response Time to FCW, Lead Vehicle Scenarios Only

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Effect	Level	Mean PABT (s)
Vehicle Model*	SRX	0.741
	XTS	0.790
	Equinox	0.731
Time of Day	Night	0.811
	Day	0.701
Setting	Medium	0.709
	Near	0.737
	Off	0.827
	Far	0.747

Large-Scale Field Test of Forward Collision Alert And Lane Departure Warning Systems, DOT HS 812 247, 2016
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812247b-lanedeparturewarningsystems.pdf

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P-AEB Inconsistent Performance

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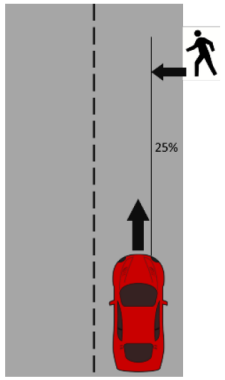
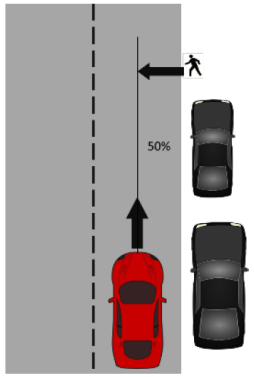
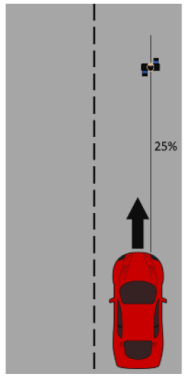
No Impact Tests / Test Conducted		
Vehicle 1	Vehicle 2	Vehicle 3
5/7 (0.8)	7/7	7/7
6/7 (15.9)	6/6	4/7 (12.6)
6/7 (7.1)	2/7 (0.4)	7/7
5/7 (22.8)	7/7	6/7 (22.4)

Objective Test Procedures for Pedestrian Automatic Emergency Braking Systems, Heath Albrecht, NHTSA & SAE, 2017
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/documents/sae2017albrecht.pdf

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IIHS Pedestrian Test Scenarios

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Parameter	*Scenario*		
	Perpendicular adult (CPNA-25)	**Perpendicular child (CPNC-50)**	**Parallel adult (CPLA-25)**
Test vehicle speed	20, 40 km/h	20, 40 km/h	40, 60 km/h
Pedestrian target speed	5 km/h	5 km/h	0 km/h
Target direction	Crossing (R-to-L)	Crossing (R-to-L)	Facing away
Target path (relative to test vehicle)	Perpendicular	Perpendicular	Parallel
Pedestrian dummy size	Adult	Child	Adult
Dummy articulation (fixed rate)	Yes	Yes	No
Overlap	25%	50%	25%
Obstructed	No	Yes	No
Number of valid runs	5	5	5
Test diagram			

Pedestrian Autonomous Emergency Braking Test Protocol (Version II), February 2019, IIHS
https://www.iihs.org/media/f6a24355-fe4b-4d71-bd19-0aab8b39aa7e/TfEBAA/Ratings/Protocols/current/test_protocol_pedestrian_aeb.pdf

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Pedestrian Targets – 4activesystems

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4activePS pedestrian static

4activeSystems produces dummies, which replicate human properties in size and shape. The child dummy represents a child at the age of 7 years. 4a dummies can be recognized by mono and stereo cameras as well as radar and infrared systems. In infrared is 50% refl ectivity, between 850nm and 910nm. There is no danger of crashes for the dummy, respectively the test vehicle, during a test.



- static EPT - Euro NCAP Pedestrian Target Static
- homogeneous distribution of radar cross section (RCS)
- visual signature - mono and stereo camera
- infrared signature (IR)
- crashable up to 60km/h
- adult weight < 4kg - child weight < 2kg

http://www.4activesystems.at/en/products/dummies/4activeps.html

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AAA Testing – Adult Crossing Path

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https://www.youtube.com/watch?v=eEnklD0Vs4A

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AAA Testing – 2019 Toyota Camry

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20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
1.255	1.079	1.229	1.330	1.339	**1.246**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
35.83	30.12	34.89	38.40	38.12	**35.47**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
30.15	26.15	30.88	28.71	29.34	**29.05**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
0.897 @ 23.81	1.024 @ 20.16	1.096 @ 23.14	0.842 @ 21.70	1.053 @ 22.41	**0.982 @ 22.24**
Impact Speed (mph)					
0.0	0.0	0.0	0.0	0.0	**0.0**
Seperation Distance At End of Test (ft)					
4.67	5.25	4.26	4.27	4.54	**4.60**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Testing – 2019 Honda Accord

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20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
0.023	1.153	0.865	1.000	0.534	**0.715**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
0.92	33.71	25.48	29.39	16.04	**21.11**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
N/A	31.71	24.32	27.62	15.15	**24.70**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
N/A	0.883 @ 5.62	0.999 @ 14.29	1.018 @ 13.90	1.950 @ 4.92	**1.213 @ 9.68**
Impact Speed (mph)					
20.1	0.0	0.0	0.0	0.6	**4.1**
Seperation Distance At End of Test (ft)					
0.00	3.91	3.57	4.10	0.00	**2.32**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Testing – 2019 Chevrolet Malibu

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20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
2.480	2.410	2.366	2.31	1.064	**2.126**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
74.12	69.54	71.71	69.50	31.53	**63.28**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
N/A	N/A	N/A	N/A	N/A	**N/A**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
N/A	N/A	N/A	N/A	N/A	**N/A**
Impact Speed (mph)					
19.0	19.5	18.3	19.7	19.7	**19.2**
Seperation Distance At End of Test (ft)					
0.00	0.00	0.00	0.00	0.00	**0.00**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Testing – 2019 Chevrolet Malibu, 30mph

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30 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
2.103	DNT	DNT	DNT	DNT	**2.103**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
93.77	DNT	DNT	DNT	DNT	**93.77**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
27.1	DNT	DNT	DNT	DNT	**27.1**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
0.776 @ 18.34	DNT	DNT	DNT	DNT	**0.776 @ 18.34**
Impact Speed (mph)					
25.5	DNT	DNT	DNT	DNT	**25.5**
Seperation Distance At End of Test (ft)					
0.00	DNT	DNT	DNT	DNT	**0.00**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Testing – 2019 Tesla Model 3

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20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
0.807	0.935	3.636	0.792	1.057	**1.445**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
23.41	27.40	105.38	22.91	29.71	**41.76**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
4.96	0.74	1.15	5.62	3.82	**3.26**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
1.083 @ 0.52	0.108 @ 0.74	0.110 @ 1.15	0.991 @ 1.67	0.868 @ 1.10	**0.632 @ 1.04**
Impact Speed (mph)					
17.3	19.6	19.0	16.8	17.5	**18.0**
Seperation Distance At End of Test (ft)					
0.00	0.00	0.00	0.00	0.00	**0.00**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Testing – Child darting out from between cars

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https://www.youtube.com/watch?v=eEnklD0Vs4A

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AAA Testing – 2019 Toyota Camry – Child darting out from between cars

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20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
N/A	N/A	N/A	N/A	DNT	**N/A**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
N/A	N/A	N/A	N/A	DNT	**N/A**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
N/A	N/A	N/A	N/A	DNT	**N/A**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
N/A	N/A	N/A	N/A	DNT	**N/A**
Impact Speed (mph)					
20.2	20.0	19.8	20.2	DNT	**20.1**
Seperation Distance At End of Test (ft)					
0.0	0.0	0.0	0.0	DNT	**0.0**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Testing – 2019 Chevrolet Malibu – Child darting out from between cars

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20 mph					
Run 1	Run 2	Run 3	Run 4	Run 5	Average
Notification TTC (s)					
1.303	0.583	1.047	1.225	0.901	**1.012**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)					
37.10	17.06	29.92	38.89	26.41	**29.88**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)					
19.01	10.09	20.63	N/A	N/A	**16.58**
Max Deceleration (G)/Associated Longitudinal Distance (ft)					
0.391 @ 13.40	0.312 @ 7.22	0.353 @ 13.67	N/A	N/A	**0.352 @ 11.43**
Impact Speed (mph)					
17.1	19.9	16.6	20.5	19.5	**18.7**
Seperation Distance At End of Test (ft)					
0.0	0.0	0.0	0.0	0.0	**0.0**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Testing – All vehicles at night

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Run 1	Run 2	Run 3	Run 4	Average
Notification TTC (s)				
N/A	N/A	N/A	N/A	**N/A**
Notification Longitudinal Distance (ft) (Vehicle to Dummy)				
N/A	N/A	N/A	N/A	**N/A**
Braking Longitudinal Distance (ft) (Vehicle to Dummy)				
N/A	N/A	N/A	N/A	**N/A**
Max Deceleration (G)/Associated Longitudinal Distance (ft)				
N/A	N/A	N/A	N/A	**N/A**
Impact Speed (mph)				
24.1	25.3	24.0	24.7	**24.5**
Seperation Distance At End of Test (ft)				
0.00	0.00	0.00	0.00	**0.00**

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA - Nighttime Pedestrian Detection

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"75% percent of pedestrian fatalities occurred in the dark
as compared to daylight (21%), dusk (2%) and dawn (2%)."

"Drivers must not rely on assistance from current pedestrian detection systems during nighttime driving or other environments with reduced visibility"

"The system must not brake for pedestrians alongside the vehicle in an adjacent sidewalk."

"Drivers must never rely on these systems to prevent a collision with a pedestrian; instead only considering them as a backup of last resort."

Automatic Emergency Braking With Pedestrian Detection, AAA, October 2019
https://www.aaa.com/AAA/common/aar/files/Research-Report-Pedestrian-Detection.pdf

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AAA Testing - Successes in Pedestrian Detection

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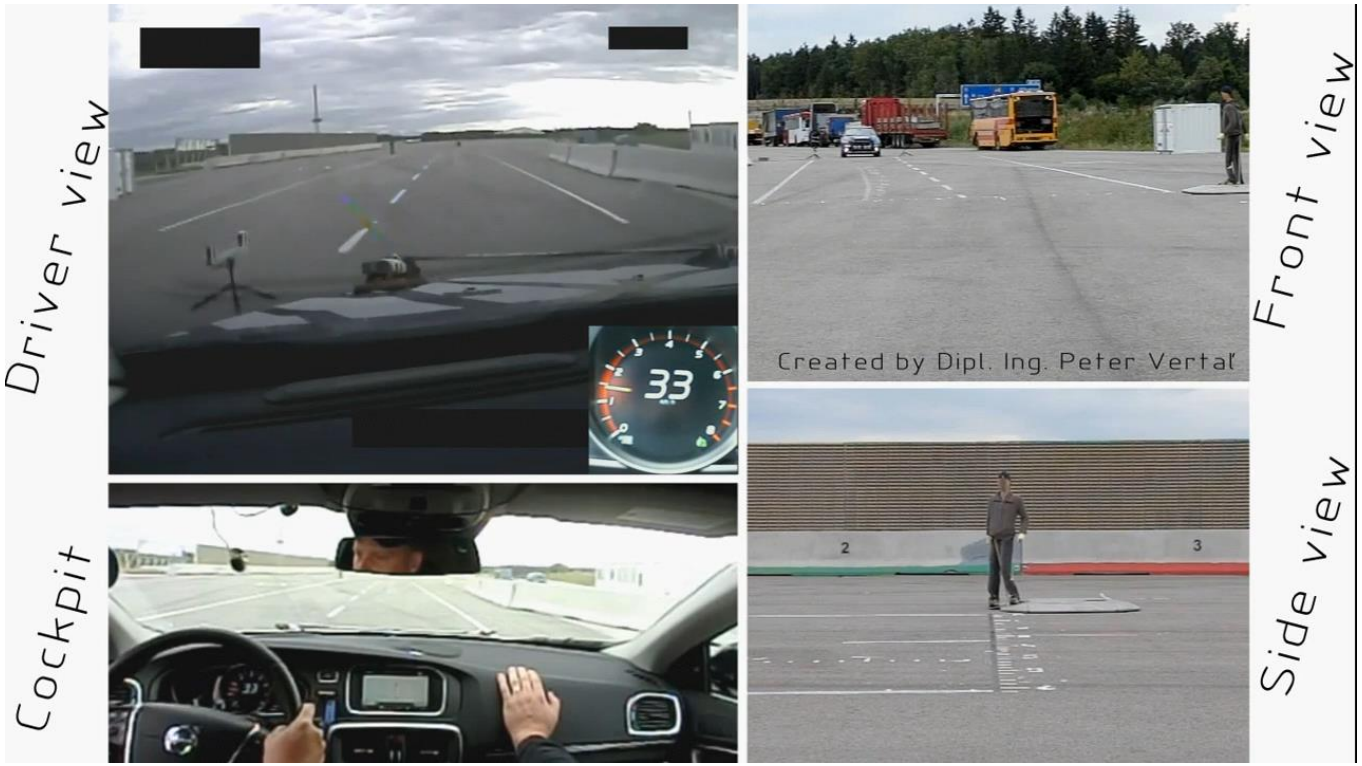


https://www.youtube.com/watch?v=eEnklD0Vs4A

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Volvo Pedestrian Detection (~2014)

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https://www.youtube.com/watch?v=XIK8L2oAkas
Volvo Pedestrian Detection, 2014 - Peter Vertal, Institute of Forensic Engineering, Brno University of Technology

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IIHS Testing – 2019 Subaru Forester

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2019 Subaru Forester

Test Speed (km/h)	TTC (s)		Max Decel (m/s^2)	Initial Speed (km/h)	Impact Speed (km/h)
	FCW	AEB			
20	0.81	0.69	9.13	20.15	0.00
20	1.00	0.85	9.06	19.85	0.00
20	0.77	0.66	11.20	20.70	0.00
20	0.89	0.74	9.61	20.09	0.00
20	0.74	0.71	11.06	20.10	0.00
40	1.57	1.39	8.73	39.25	0.00
40	1.50	1.40	9.62	39.90	0.00
40	1.49	1.31	10.23	39.95	0.00
40	1.61	1.49	10.81	39.45	0.00
40	1.47	1.36	9.19	40.16	0.00
20	1.06	1.00	7.27	20.53	0.00
20	1.18	1.03	9.52	19.98	0.00
20	1.05	0.98	8.77	20.52	0.00
20	1.11	0.97	8.50	19.75	0.00
20	1.06	0.87	8.61	20.50	0.00
40	1.08	0.93	11.14	39.47	0.00
40	1.08	0.97	7.97	40.05	0.00
40	1.01	0.86	10.38	40.11	0.00
40	1.08	0.96	9.90	39.95	0.00
40	0.95	0.79	9.63	40.11	0.00
Average	**1.13**	**1.01**			

Adult Crossing Scenarios

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IIHS Testing – 2018 BMW X1

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2018 BMW X1

Test Speed	TTC (s)		Impact Speed (km/h)
	FCW	AEB	
20	0.08	0.02	20.30
20	0.00	0.00	20.71
20	0.00	0.00	19.59
20	0.14	0.10	19.73
20	0.14	0.19	18.83
40	0.44	0.35	36.66
40	0.47	0.44	35.90
40	0.46	0.45	35.16
40	0.48	0.44	35.75
40	0.32	0.39	35.87
20	0.49	0.16	18.37
20	0.00	0.00	20.38
20	0.59	0.60	12.26
20	0.49	0.00	19.95
20	0.81	0.72	8.96
40	0.53	0.45	35.71
40	0.66	0.52	34.50
40	0.49	0.35	36.68
40	0.47	0.51	34.65
40	0.57	0.63	32.26
Average	**0.38**	**0.32**	

Adult Crossing Scenarios

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Volvo P-AEB Failure at Car Dealership

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Forensic Engineering



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Test variables to watch for, in both published and exemplar testing

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Forensic Engineering

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- Time of day and lighting
- Background objects and visual noise
- Driver inputs, which can inhibit systems
- Biofidelic motion
- Contrast
- Structure from motion

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Forensic Engineering

How to reconstruct the P-AEB crash

2020 Collision Publishing

Forensic Approach to Pedestrian AEB-equipped Accidents

a.b.moore Forensic Engineering

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Investigation: Was the vehicle equipped with ADAS technology?

Analysis: Did it perform as designed?

Effectiveness: Could it have mitigated or prevented the accident?

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P-AEB EDR - Toyota

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Vehicle Control History via Techstream can provide:

- Target type
- Target speed toward vehicle, and across path
- Target distance
- Warnings and autobrake triggers
- Driver response

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P-AEB EDR – General Motors

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CDR 900 can provide:

- Some pedestrian classification
- Target speed toward vehicle, and across path
- Target distance
- Warnings and autobrake triggers
- Driver response
- Override/Inhibit Reasons

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GM Active Safety Control Module (ASCM)

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Automated System Control Data, -3.92 to 0.00 sec - Table 5 of 7

Time (sec)	FPB Suppression Reason: Out of Range or Above Time to Collision Threshold	FPB Suppression Reason: Pedestrian Not Seen	FCA/FPB Suppression Reason: Accelerator or Clutch Pedal Applied	FCA/ FPB Suppression Reason: Brake Pedal Applied	FCA Suppression Reason: Above Time to Collision Threshold	FCA Suppression Reason: Out of Range or Moving Away

2017 Cadillac CTS-V

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GM Active Safety Control Module (ASCM)

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Automated System Control Data, -3.92 to 0.00 sec - Table 6 of 7

Time (sec)	FCA Suppression Reason: Imminent Alert Open Scenario	FCA Suppression Reason: Speed too Low or Hi Threat Pedestrian	Headway Suppression Reason: High Range, Low Speed, or Opening Scenario	FCA/Headway Suppression Reason: Vehicle out of Path	FCA/Headway Suppression Reason: Oncoming Vehicle	FCA Suppression Reason: Never Before Seen Moving Object and Low Host Speed

2017 Cadillac CTS-V

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Subaru Eyesight EDR

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Japan-spec vehicle, Subaru Levorg, with Eyesight 3

"Subaru EyeSight Version 3 can only be accessed by Subaru personnel with special software. Subaru's current policy on assisting in data retrieval is only to assist proper legal authority with written owner consent."



Accident Reconstruction with Data Recorded by Electronic Control Units in Vehicles with a Pre-crash Safety System, Oga, et al, SAE 2018-01-1440

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